CS 331 – Software Engineering Lab

Assignment 6 – User Interface Design

I. Choice of User Interface and Justification

For the InsightFlow system, we implemented a **Menu-Based Web Interface with Direct Manipulation elements**.

The user interface is implemented as a **browser-based application** that allows users to interact with the system through graphical components such as buttons, input fields, and navigation menus.

The UI provides the following features:

- User **registration and login**
- Upload **CSV files for report generation**
- Upload **PDF files for document analysis**
- **Chat with PDF** functionality
- Generate **automated reports**

Justification for choosing this UI

1. Ease of Use

A web-based menu interface is easy for users to understand and operate without technical knowledge.

2. Supports File Upload

Since the application requires uploading **PDF and CSV files**, a graphical interface with file selectors is more suitable than command-line interfaces.

3. Direct Interaction

Users can directly interact with UI elements such as:

- Upload buttons
- Chat input box
- Login / signup forms
- Report generation buttons

4. Accessibility

The application runs in a web browser, allowing users to access the system from any device without installing additional software.

5. Reduced Errors

Menu-based systems reduce the chances of input errors because users interact with predefined UI elements rather than typing commands manually.

Therefore, a **menu-based web interface combined with direct manipulation elements** is the most suitable UI for InsightFlow.

II. Implementation of UI Components and User Interaction

The InsightFlow interface is implemented using **HTML, CSS, and JavaScript** and communicates with backend microservices using REST APIs.

Below are the key UI components implemented in the system.

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1. User Registration Interface

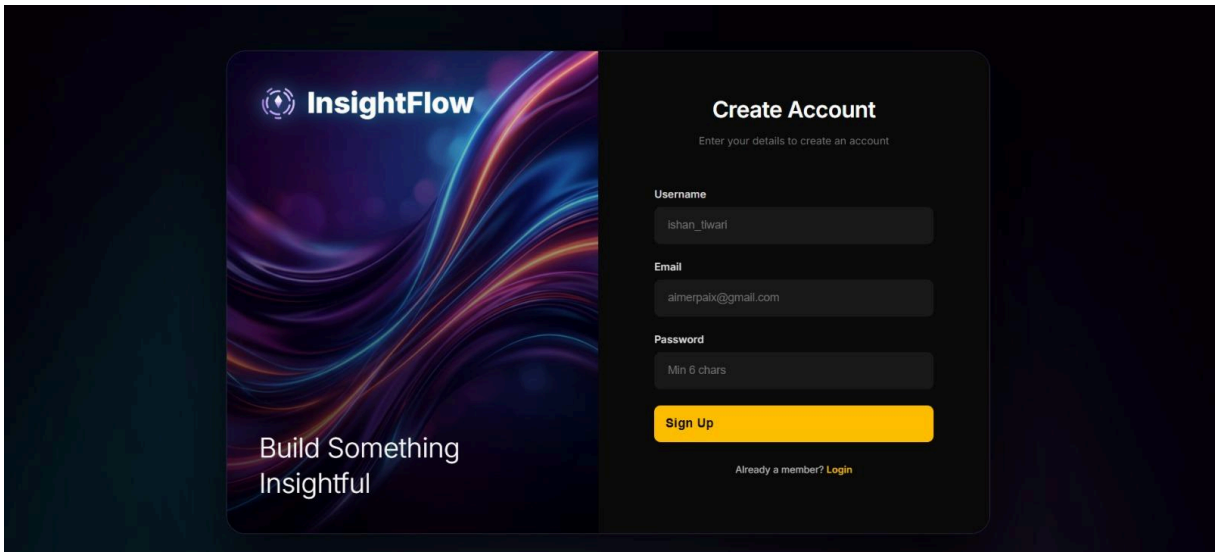
The system provides a **signup page** where new users can create an account by entering:

- Username
- Email
- Password

Once the user submits the form, the frontend sends the data to the backend authentication API.

User Interaction

User → Signup form → Backend API → Account created



Description

The user registration interface allows new users to create an account in the InsightFlow system.
Users enter their **username, email, and password** and click the **Create Account** button.

Interaction Flow

User → Signup Form → Frontend sends request → Authentication API → Account created.

2. User Login Interface

Registered users can log in using their credentials.

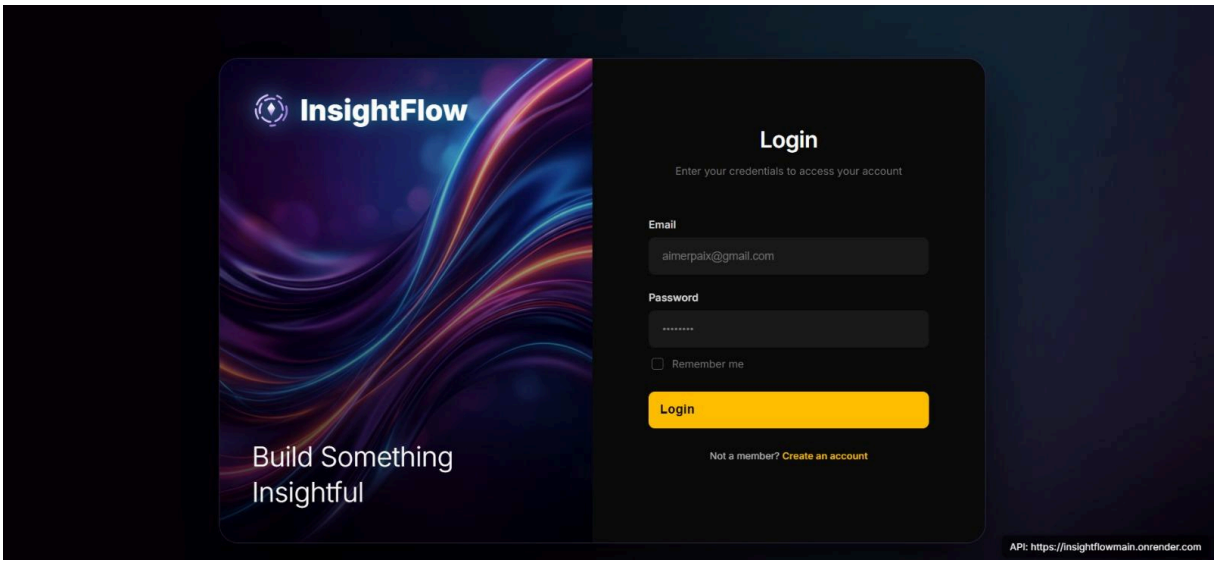
The login interface contains:

- Email field
- Password field
- Sign-in button

After successful authentication, the user is redirected to the main dashboard.

User Interaction

User → Login form → Backend authentication → Access dashboard



Description

The login interface allows existing users to access the InsightFlow system using their registered credentials.
Users provide their **email and password** and click the **Sign In** button to access the application dashboard.

Interaction Flow

User → Login Form → Backend Authentication Service → User redirected to dashboard.

3. InsightFlow Dashboard Interface

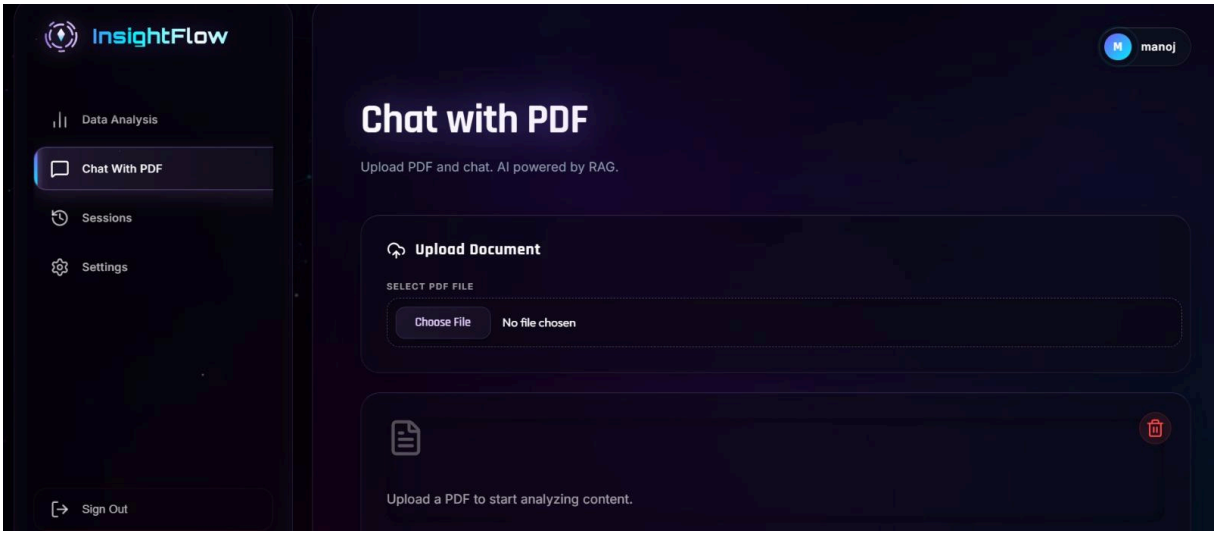
After login, users are directed to the main application interface where they can access the core features of the system.

The dashboard includes navigation options such as:

- Upload CSV
- Upload PDF
- Chat with document
- Generate reports
- View system features

User Interaction

User → Dashboard → Select feature → Frontend sends request to backend services



Description

The InsightFlow dashboard provides access to the main features of the system.

From this interface, users can navigate to different functionalities such as:

- Upload CSV files for report generation
- Upload PDF documents for analysis
- Chat with uploaded documents
- Generate automated reports

The dashboard acts as the central control interface for interacting with the system.